THE EVALUATION MAP IDENTIFIES R-HOM(L,M) TENSOR K WITH R-HOM(R-HOM(K,L), M) FOR PERFECT K

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Behaviour of the internal hom under restriction to opens. Stacks Project, Cohomology of Sheaves, Lemma lemma-restriction-RHom-to-U (tag 0A8M). Justifies the reduction to a local computation on X.
- The evaluation map and its basic compatibility. Stacks Project, Cohomology of Sheaves, Lemma lemma-internal-hom-evaluate (tag 0A8N). Provides the comparison map whose bijectivity is being proved.

Source reference. Original lemma 0A8U: sources/stacks-chow/tags.txt, source lines 12927–12927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DL: sources/stacks-chow/tags.txt, source lines 10641–10641. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original section 06Y7: sources/stacks-chow/tags.txt, source lines 8876–8876. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08I0: sources/stacks-chow/tags.txt, source lines 10798–10798. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 014A: sources/stacks-chow/tags.txt, source lines 1465–1465. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $K, L, M \in D(\mathcal{O}_X)$. If K is perfect, then the map*

$$R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K \longrightarrow R\mathcal{H}om(R\mathcal{H}om(K, L), M)$$

of Lemma 0A8U is an isomorphism.

Proof. Since the map is globally defined and since formation of the right and left hand side commute with localization (see Lemma 08DL), to prove this we may work locally on X. Thus we may assume K is represented by a strictly perfect complex $\mathcal{E}^\bullet$.

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If $K_1 \to K_2 \to K_3$ is a distinguished triangle in $D(\mathcal{O}_X)$, then we get distinguished triangles

$$R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K_1 \to R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K_2 \to R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K_3$$

and

$$R\mathcal{H}om(R\mathcal{H}om(K_1, L), M) \to R\mathcal{H}om(R\mathcal{H}om(K_2, L), M) R\mathcal{H}om(R\mathcal{H}om(K_3, L), M)$$

See Section 06Y7 and Lemma 08I0. The arrow of Lemma 0A8U is functorial in K hence we get a morphism between these distinguished triangles. Thus, if the result holds for K_1 and K_3, then the result holds for K_2 by Derived Categories, Lemma 014A.

Combining the remarks above with the distinguished triangles

$$\sigma_{\geq n}\mathcal{E}^\bullet \to \mathcal{E}^\bullet \to \sigma_{\leq n-1}\mathcal{E}^\bullet$$

of stupid trunctions, we reduce to the case where K consists of a direct summand of a finite free $\mathcal{O}_X$-module placed in some degree. By an obvious compatibility of the problem with direct sums (similar to what was said above) and shifts this reduces us to the case where $K = \mathcal{O}_X^{\oplus n}$ for some integer n. This case is clear. □